27

25

1. $\dfrac{4\pi^2 kR}{4\pi^2 k - M\omega^2}$（6

2. $\dfrac{fV}{V+u}$，$\dfrac{fV}{V-u}$（4

3. $T_0\left(1-\dfrac{Mg}{p_0 A}\right)^{\frac{1-1}{\gamma}}$（6

4. $7.3\times10^6\,\mathrm{m\cdot s^{-1}}$，$2.8\times10^8\,\mathrm{m\cdot s^{-1}}$，$8.4\times10^5\,\mathrm{V}$（9

$$p=\frac{h}{\lambda}$$

$$E=\sqrt{p^2c^2+m_e^2 y}$$

$$E_k=eV=E-m_e c^2$$

20

1. m

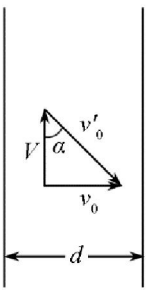
$$a=\frac{mg\sin\theta-\mu_1 mg\cos\theta}{m}=g\left(\sin\theta-\mu_1\cos\theta\right),\qquad(1)$$

$$v_0=\sqrt{\frac{2ah}{\sin\theta}}=\sqrt{2gh\left(1-\mu_1\cot\theta\right)}=4.0\mathrm{m/s}\qquad(2)$$

$$v_0'=\sqrt{v_0^2+V^2}=5.0\mathrm{m/s}\text{。}\qquad(3)$$

α

$$\tan\alpha=\frac{4}{3}\text{。}\qquad(4)$$



$$a'=\frac{\mu_2 mg}{m}=\mu_2 g\text{。}\qquad(5)$$

$$s'=\frac{v_0'^2}{2a}=\frac{v_0'^2}{2\mu_1 g},\qquad(6)$$

μ_2